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FLOTATION PROCESS

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This invention relates to a new flotation process and more particularly refers to a froth flotation process for concentrating carbonaceous materials.

Heretofore large quantities of carbonaceous material, in particular bituminous and anthracite coal, have been discarded because they contained deleterious constituents such as slate which prevented their use as fuel. Considerable time and money has been spent in attempting to separate these deleterious constituents from the valuable carbonaceous residue. These attempts have, however, in great part been unsuccessful because of the appreciable expense which was involved and the substantial failure to obtain a satisfactory separation of the desired constituents from the contaminating products. It has been proposed to concentrate the desired carbonaceous material by flotation processes. However, due to the fact that the deleterious constituents had approximately the same floatability as the desired carbonaceous material such proposed processes were in large part unsuccessful.

It is an object of the present invention to overcome the aforementioned defects as well as additional defects which have been described in the literature. A further object is to separate carbonaceous materials from deleterious inorganic and other constituents with which they are associated by means of a process which is both cheap and efficient. A still further object is to concentrate carbonaceous material by means of a froth flotation process which has greater selectivity than any heretofore devised. A still further object is to concentrate coal and thereby separate it from slate and other contaminants by a froth flotation process which is applicable over a wide range of conditions and which may utilize numerous flotation agents. A still further object is to utilize the vast quantities of low grade bituminous and anthracite coal which have heretofore been discarded, and to obtain therefrom bituminous and anthracite coal of high quality. Additional objects will become apparent from a consideration of the following description.

These objects are attained according to the herein described invention which comprises agitating the carbonaceous material in a liquid medium to which has been added a sulfonated hydrocarbon. In its preferred embodiment the invention comprises agitating coal, in particular anthracite coal, in an aqueous medium to which has been added an aqueous emulsion comprising a water-soluble sulfate salt of an alcohol

containing from 6 to 18 carbon atoms and a water-repellent organic compound. For optimum results the preformed aqueous emulsion should in general contain a water-soluble sulfate salt of a normal primary alcohol having from 12 to 18 carbon atoms and a hydrocarbon or mixture of hydrocarbons having water-repellent properties.

The invention may be more readily understood by a consideration of the following illustrative example:

Example

300 cc. of an emulsion was prepared by mixing the following solutions:

- (a) 0.263 gram of the sodium sulfate salt of technical oleyl alcohol
- 2.00 grams of rosin FF
- 2.50 grams of sodium hydroxide
- 100 cc. of water

The mixture was heated to boiling to dissolve the reagents.

- (b) 15 grams of crude petroleum oil
- 2 grams of rosin oil residue

The mixture was warmed to about 50° C.

Solution (a) was added to mixture (b) and an emulsion prepared by thoroughly stirring the product. This emulsion was extended in 900 cc. of water.

The aforementioned emulsion was added to a flotation cell containing an aqueous suspension of 1025 grams of anthracite coal from the Susquehanna Colliery Co., Shamokin, Pa. The ash content of this coal was 17.09%.

During flotation 5 drops of pine oil (0.10 gram) were added.

The pH value of the suspension was maintained at about 7.0.

422 grams of concentrate were obtained assaying 11.34% ash.

537 grams of low grade concentrate were obtained assaying 15.06% ash.

The tailings amounted to 61 grams and assayed 69.12% ash.

By repeating the above experiment using a larger quantity of the sodium sulfate salt of oleyl alcohol it was found unnecessary to add any pine oil during the flotation operation.

The above experiment was repeated using water-soluble sulfate esters of alcohols, both saturated and unsaturated, having from 6 to 18 carbon atoms in the molecule, as well as mixtures thereof. In each test it was found that a high grade concentrate was obtained, the ma-

jority of slate and other deleterious constituents being confined to the tailings.

It is to be understood that the above examples are illustrative merely of the present invention, and are not intended as a limitation thereon. The carbonaceous material floated may be selected from a wide variety of sources. It may be, for example, either bituminous or anthracite coal and may be associated with deleterious material of a wide variety such as inorganic ash-forming constituents. Other carbonaceous material may also be used, the invention not being limited to the flotation of bituminous or anthracite coal.

The sulfonated hydrocarbon may be obtained from various sources. It is preferably obtained from a normal primary alcohol having from 12 to 18 carbon atoms, but may also be derived from any alcohol or alcohol derivative containing from 6 to 18 carbon atoms. This is understood to include not only the open chain compounds but also cyclic compounds such as naphthenic alcohols. The alcohols may be saturated or unsaturated, and may also be primary, secondary and/or tertiary. In selecting alcohols or equivalent compounds having less than 12 carbon atoms it is frequently advisable to choose those compounds which contain side chains. Likewise, the presence of small amounts of ketones and/or aldehydes is in general advantageous. The latter assistants may be illustrated by acetone, formaldehyde, acetaldehyde, etc. While alcohols are frequently referred to herein as the source of the sulfonated derivatives it is contemplated that similar derivatives may be obtained from hydrocarbons or their substituted derivatives which are non-alcoholic compounds according to well known reactions. An example of such optional process is the sulfonation of olefines and diolefines.

The sources of these alcohols and related compounds are well known and need not be described in detail herein. However, mention may be made of the fact that animal, fish and vegetable oils, fats and waxes are particularly adapted for this purpose. For instance, sperm oil or herring oil may be saponified with lime or other basic material and the resulting alcohols, primarily oleyl, cetyl and stearyl alcohols, separated therefrom by fractional distillation. Likewise, the natural compounds previously mentioned and other compounds suggested thereby, including synthetically prepared compounds, may be converted to alcohols by the sodium reduction or catalytic hydrogenation method. Representative sources of materials contemplated herein are coconut oil, palm kernel oil, beeswax, carnauba wax, Montan wax, lanolin, wool fat, spermaceti, sperm oil, herring oil, olive oil, castor oil, linseed oil, etc. A few of the alcohols derived therefrom are, for example, octyl, decyl, lauryl, myristyl, cetyl, stearyl, oleyl, ricinoleyl, linoleyl, etc. Other hydrocarbon derivatives contemplated herein may be obtained by the catalytic reduction of carbon monoxide with hydrogen.

Sulfonation of the alcohols contemplated herein or related compounds may advantageously be accomplished by treatment with well known sulfonating agents, such as chlorosulfonic acid, sulfuric acid or oleum. This treatment may be carried out in the presence of organic solvents or diluents such as nitrobenzene, carbon tetrachloride, etc. It may also be carried out in the presence of agents having water-binding properties as, for example, acetic anhydride, sulfur trioxide, phthal-

ic anhydride and other well known acid anhydrides or anhydrous acids.

For optimum results sulfate esters of the hydrocarbons are in general preferred. These esters may readily be prepared by conducting the aforementioned sulfonation treatment at reduced temperatures such as temperatures within the range of 0-30° C. The invention is, however, not confined to sulfate esters since true sulfonates may also be used in place thereof or in admixture therewith. In this connection, mention may be made of the fact that where sulfonates are referred to broadly herein it is intended that such term include both true sulfonates and sulfate esters.

The sulfonated hydrocarbons produced, for example in accordance with the above instructions, are advisably neutralized by treatment with salt-forming compounds. These compounds may be of organic or inorganic origin. Compounds coming within this category are caustic soda, soda ash, caustic potash, ammonium hydroxide, triethanolamine, pyridine, piperidine, cyclohexylamine, mono- and dialkyl cyclohexylamine, calcium, magnesium, lithium, aluminum, lead, and barium salts, etc. The aforementioned sulfonated hydrocarbons may be represented by the following general formula:



wherein R represents a radical containing a hydrocarbon group of at least 6 carbon atoms, X represents a sulfuric acid or sulfonic acid group, and Y represents a salt forming group. It is to be understood that the use of water-soluble sulfate salts of normal primary alcohols having from 12 to 18 carbon atoms in the molecule is preferred herein for the most satisfactory results over a wide range of conditions. These preferred compounds conform to the following general formula:



wherein R represents the residue of a normal primary alcohol containing from 12 to 18 carbon atoms and Y represents the residue of a salt forming compound. As is clear from a consideration of the previous description, the invention is broader in its scope than the aforesaid preferred embodiment. Water-insoluble salts are likewise included herein but when such salts are selected it is ordinarily helpful to dissolve or disperse them in organic solvents or dispersing agents prior to their addition to the water-repellent substances.

Water-repellent compounds are preferably hydrocarbons such as crude oil or fractions of petroleum having a relatively high molecular weight. The selection of such compounds is not limited to the petroleum field, since synthetic or naturally occurring derivatives of hydrocarbons may also be used in place thereof or in admixture therewith. The hydrocarbon derivative which is used should advisably not have too powerful a collecting power since the deleterious constituents may then be concentrated with the carbonaceous material. The primary requisite of the hydrocarbon derivative which is selected for this purpose is that it be water-repellent and that it have a relatively moderate collecting power.

Most satisfactory results are customarily obtained by emulsifying the water repellent agent prior to its addition to the flotation cell. This emulsification is advisably accomplished by means of a salt of a sulfonated normal primary alcohol, in accordance with the instructions given supra. The invention is to be understood as contemplated

ing the addition of these compounds to the vessel for disintegrating the carbonaceous material or the flotation cell in unemulsified form, the emulsion being formed in situ. The results obtained when this latter method is used are in general inferior to those obtained when the emulsion is made prior to addition of the described compounds to the flotation cell.

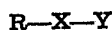
It is to be understood that the present invention is capable of use in conjunction with numerous well known emulsifying and flotation agents. For example, it may contain free alcohols, preferably, but not exclusively, those corresponding in number of carbon atoms to the sulfonated hydrocarbon used or homologues thereof. It may also contain assistants such as soap, rosin, pine and hardwood oils, creosote, cresylic acid, sulfonated higher fatty acids such as oleic and/or ricinoleic acids, Twitchell's reagent, naphthalene-sulfonic acids, polyethanolamines and their derivatives such as soaps prepared from triethanolamine and higher fatty acids, e. g. oleic, palmitic and/or stearic acids, substituted ethylene diamines, for example the hydrochlorides, resinsates, sulfonated rosin, sulfonated abietene, abietic acid and its salts, condensation products of abietene or abietane with epichlorhydrin which are sulfonated, naphthenic acids, their salts and sulfonation products, etc. It will be noted that many of the aforementioned products are sulfonated hydrocarbons within the meaning of the present invention. It is therefore to be understood that while when such compounds are used they are preferably mixed with the preferred sulfonated hydrocarbons the invention is intended to include their use in place of such preferred derivatives.

The present invention renders available enormous quantities of carbonaceous materials which formerly were unsuitable because of the large proportion of inorganic compounds which were mixed therewith. The process described herein is highly selective and may be operated at surprisingly low cost. It permits concentrates of low ash content to be obtained. Likewise, the sulfonated hydrocarbons and other compounds suitable for use herein are readily available and may be used in such small quantities that the operation is highly satisfactory from a commercial standpoint.

As many apparently widely different embodiments of this invention may be made without departing from the spirit and scope thereof, it is to be understood that the invention is not limited to the specific embodiments thereof except as defined in the appended claims.

We claim:

1. A process for the froth flotation of coal which comprises agitating and aerating an aqueous suspension of coal in the presence of a mineral oil which has been emulsified with a compound having the following general formula:



wherein R represents the residue of a normal primary alcohol containing from 6 to 18 carbon atoms, X represents a sulfuric acid or sulfonic acid group, and Y represents a salt-forming compound.

2. A process for the froth flotation of coal which comprises agitating and aerating an aqueous suspension of coal in the presence of a mineral oil which has been pre-emulsified with a water-soluble compound having the following general formula:



wherein R represents the residue of a normal primary alcohol containing from 12 to 18 carbon atoms, and Y represents the residue of a salt-forming compound.

3. The process of claim 2 wherein flotation is carried out in the presence of rosin, in addition to the agents defined therein.

4. A process for the froth flotation of anthracite coal which comprises agitating and aerating an aqueous suspension of anthracite coal in the presence of a mineral oil which has been pre-emulsified with a water-soluble compound having the following general formula:



wherein R represents the residue of a normal primary alcohol containing from 12 to 18 carbon atoms, and Y represents the residue of an organic amine salt-forming compound.

5. A process for the froth flotation of anthracite coal which comprises agitating and aerating an aqueous suspension of anthracite coal in the presence of a mineral oil which has been pre-emulsified with a water-soluble compound having the following general formula:



wherein R represents the residue of a normal primary alcohol containing from 12 to 18 carbon atoms, and Y represents the residue of a cyclohexylamine.

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